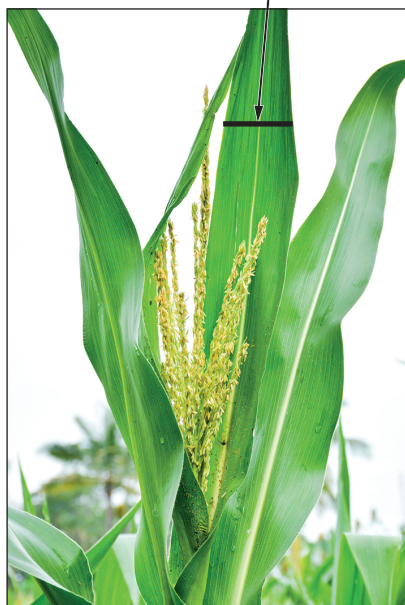


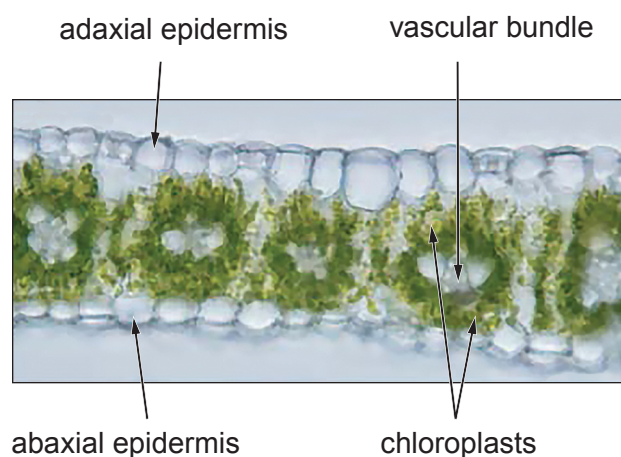
2. The distribution of chloroplasts inside leaves is important for efficient photosynthesis. Millet is a grass-like plant in which the leaves grow upwards so that they have an inward-facing (adaxial) surface and an outward-facing (abaxial) surface. **Image 2.1** shows a millet plant and **Image 2.2** shows a transverse section of a millet leaf.

**Image 2.1**

position of section shown in **Image 2.2**



**Image 2.2**



- (a) (i) Compare the distribution of chloroplasts in the millet leaf with the distribution of chloroplasts in a typical leaf such as *Ligustrum*. [2]

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- (ii) With reference to **Images 2.1** and **2.2**, conclude how the distribution of chloroplasts enables more efficient light absorption in millet plants. [2]

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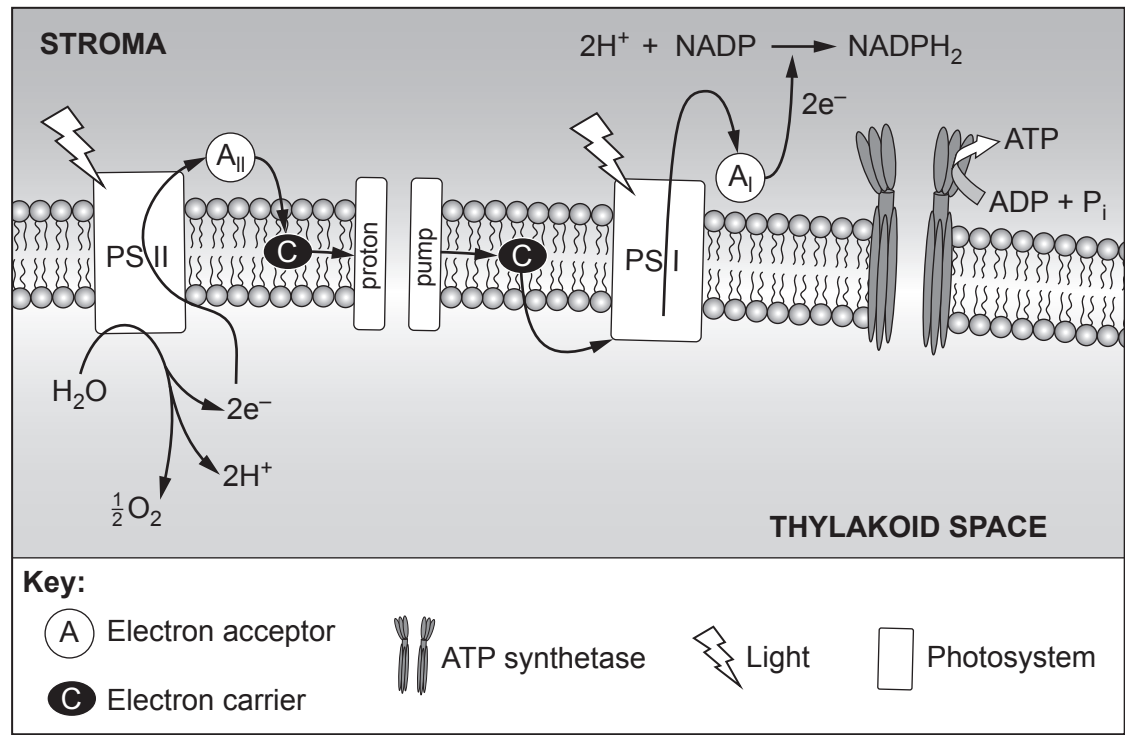
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**Image 2.3** shows how electrons pass between components of thylakoid membranes during non-cyclic photophosphorylation.

**Image 2.3**



- (b) (i) Name the process shown in **Image 2.3** by which electrons are released from water.

[1]

The relative energy levels of the electrons in some of the components in **Image 2.3** are shown in **Table 2.4**.

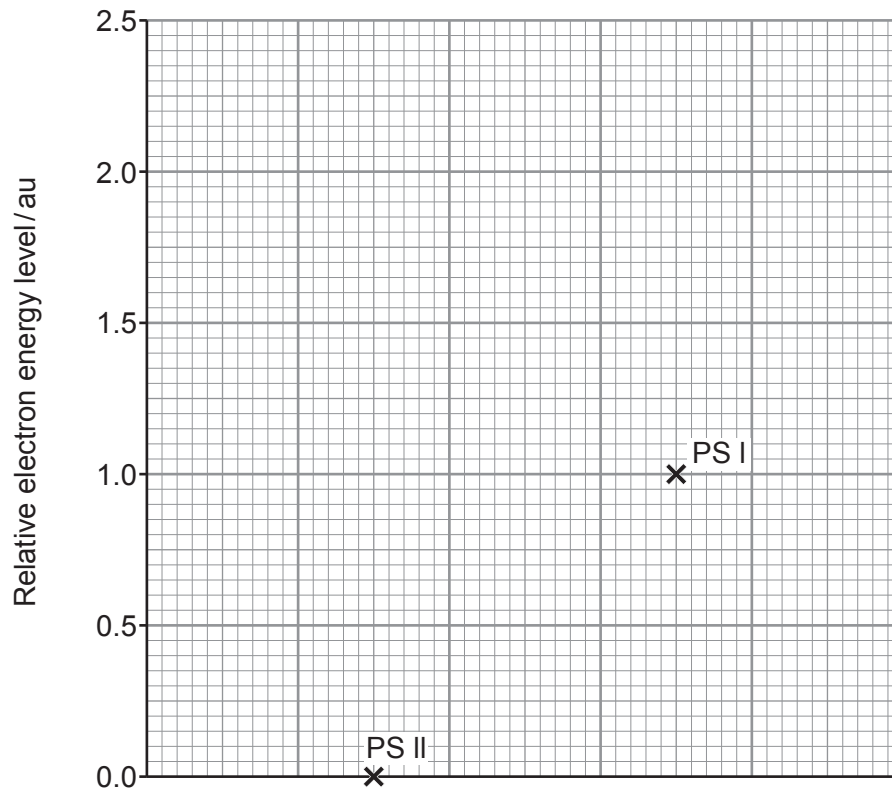
**Table 2.4**

| Membrane component         | Relative electron energy level / au |
|----------------------------|-------------------------------------|
| Photosystem II (PSII)      | 0                                   |
| Electron acceptor $A_{II}$ | 1.8                                 |
| Photosystem I (PSI)        | 1.0                                 |
| Electron acceptor $A_I$    | 2.0                                 |



- (ii) **Plot and label** the positions of the **electron acceptors A<sub>I</sub> and A<sub>II</sub>** on **Graph 2.5** (PSI and PSII have been done for you) and **join the four components with three straight lines** to show the **energy profile of non-cyclic photophosphorylation**. [2]

**Graph 2.5**



- (iii) Use your knowledge of non-cyclic photophosphorylation to describe and explain the changes in relative electron energy levels shown in **Graph 2.5**. [5]

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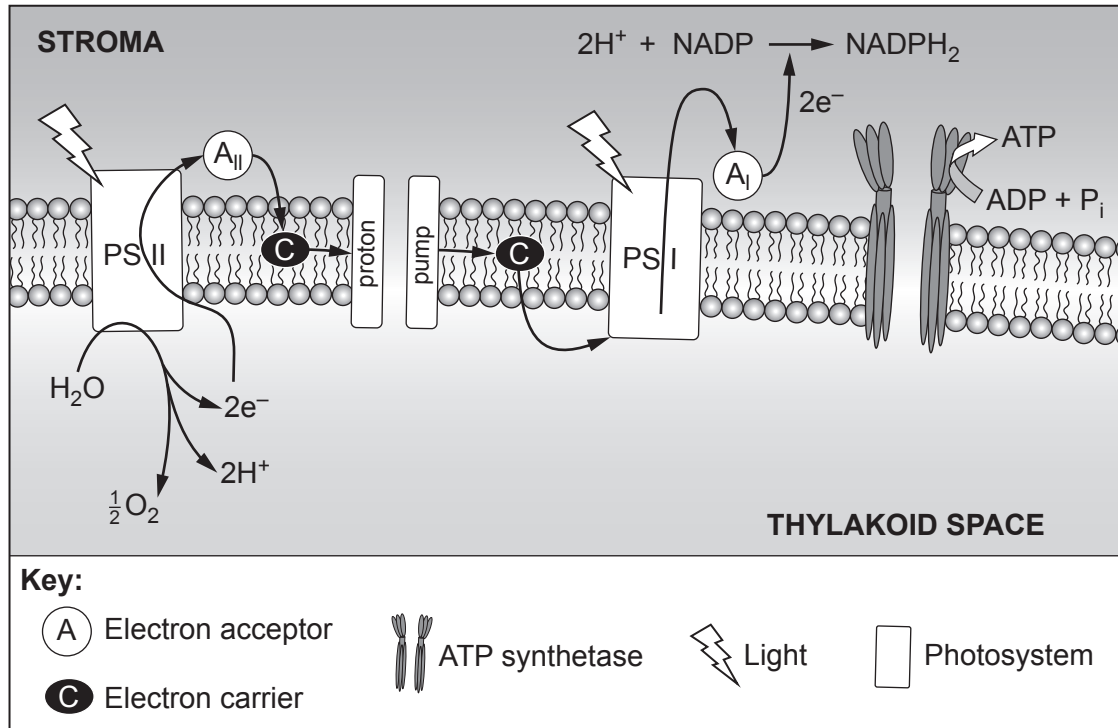
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ATP production is carried out in chloroplasts by chemiosmosis which involves creating a proton gradient across the thylakoid membrane.

**Image 2.6**



- (c) (i) **On Image 2.6:**  
**draw an arrow labelled P** through the proton pump to show the direction in which protons are pumped;  
**draw an arrow labelled D** through the ATP synthetase molecule to show the direction in which protons diffuse to activate ATP synthetase. [1]
- (ii) In addition to proton pumps, two other processes shown in **Image 2.6** also contribute to creating a proton gradient across the thylakoid membrane. Describe these **two** processes and state where each occurs. [3]

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